Vansh Singh

Computer Engineer | Cloud, Quantum ML & Embedded Systems

Dual-degree student at UMass Amherst (BS Computer Engineering '26 & MS Business Analytics '27). Experienced in automated cloud infrastructure (Terraform/AWS), embedded mechatronics, and quantum variational machine learning algorithms.

GitHub: github.com/vsingh2005 ↗

Email: vanshsingh@umass.edu

Location: Amherst, MA & Chicago, IL

2nd Place GLOBAL ASME IAM3D 40+ Int'l Universities	**40%** PROVISIONING BOOST Terraform & AWS IaC	**150+** STUDENTS TAUGHT Embedded & Firmware Labs	**28%** CONVERGENCE SPEEDUP PennyLane + AWS Braket

PROFESSIONAL EXPERIENCE

INDUSTRY & LEADERSHIP

Cloud Infrastructure Engineering Intern • University of Massachusetts Amherst IT

Sept 2025 – Present | Amherst, MA

- Architect and deploy automated cloud infrastructure and CI/CD pipelines across AWS using **Terraform**, reducing environment provisioning turnaround by **40%** and eliminating manual configuration drift.
- Develop telemetry, health-check, and monitoring services in **Python and Bash** to isolate platform faults and improve service uptime across university IT platforms.
- Write reusable Terraform modules and containerized microservices in **Docker** to standardize deployment setups across 6 engineering teams.

`AWS (EC2, S3, Lambda, IAM)` `Terraform` `Docker` `Python` `Bash` `CI/CD` `Linux/Unix`

Systems & Embedded Engineering Lead / Instructor • Stem Studio

May 2023 – Present | Chicago, IL

- Led emergency incident remediation for a server-side cloaking exploit and backdoor on production servers, restoring full system stability and uptime with zero data loss.
- Created and taught hands-on embedded computing curricula (microcontrollers, circuit design, sensor interfacing, and robotics) for **150+ students**, raising project pass rates by **35%**.
- Authored reproducible laboratory guides, hardware documentation, and sample firmware repositories.

`Embedded C` `ARM Cortex` `ESP32 / Arduino` `I2C / SPI / UART` `Linux Hardening` `Incident Response`

Technical Mentor & Lead Instructor • CodeDay

Aug 2020 – Aug 2023 | Global Initiative

- Mentored **20+ student engineering teams** through complete project lifecycles, covering data structures, API integrations, algorithms, and Git collaboration.
- Conducted regular code reviews, debugging walkthroughs, and architecture reviews to help teams deploy functional software.

`Python` `JavaScript` `REST APIs` `Git / GitHub` `Data Structures`

EDUCATION & ACADEMIC LEADERSHIP

UMASS AMHERST

Bachelor of Science in Computer Engineering • University of Massachusetts Amherst

Graduation: May 2026 | Amherst, MA

College: Riccio College of Engineering | **Subteam Lead:** ASME IAM3D Mechatronics & U.S. DOE Collegiate Wind

Key Courses: Computer Architecture, Embedded Systems Design, SystemVerilog & Digital Logic, Data Structures & Algorithms, Signals & Systems, Power Electronics.

Master of Science in Business Analytics (MSBA) • University of Massachusetts Amherst

Graduation: May 2027 | Amherst, MA

School: Isenberg School of Management | Focus on quantitative modeling, high-volume processing, and enterprise data architectures.

TECHNICAL COMPETENCIES

CORE SKILLS MATRIX

Programming & Systems
Languages: Python, JavaScript (ES6+), SystemVerilog, SQL, Java, Bash/Shell, C/C++
Quantum: Qiskit, PennyLane, Circuit Optimization, State-Vector Simulation

Embedded & Hardware
Firmware & MCUs: Embedded C, ARM Cortex, ESP32, Arduino, I2C / SPI / UART
EDA & Prototyping: LTSpice Simulation, SolidWorks CAD, CNC Machining, 3D Printing

Cloud & Infrastructure
Cloud Platforms: AWS (EC2, S3, Lambda, IAM, Braket), Google Cloud (GCP)
DevOps & Tools: Terraform (IaC), Docker, CI/CD Pipelines, Linux/Unix Internals

AI, ML & Analytics
Deep Learning: PyTorch, Variational Quantum Circuits (QNN), CNNs, NumPy
Analytics: High-Volume Processing, Predictive Modeling, REST APIs

Engineering Case Studies & Repositories

Architectural Overviews, Hardware Prototypes & Live GitHub Repositories

GitHub Hub: github.com/vsingh2005 ↗

Click any badge to view code repositories

Quantum-Classical Hybrid Neural Network

GitHub Code ↗

Variational Quantum Circuits on AWS Braket with PennyLane • Lead Architect & Developer

Designed and implemented a hybrid neural network integrating classical convolutional layers with parameterized variational quantum circuits executed on AWS Braket state-vector simulators, achieving 28% faster epoch convergence and 3.4x parameter efficiency.

28% Faster	3.4x	AWS Braket
Epoch Convergence vs Classical CNN	Parameter Efficiency in Hilbert Space	Distributed Quantum Simulation

Key Highlights: Custom PennyLane QNode integrated with PyTorch `nn.Module` forward loops; angle embedding with multi-qubit entanglement (CNOT + LCU); parameter-shift gradient computation; gate depth optimization.

`PyTorch` `PennyLane` `AWS Braket` `Python` `Qiskit` `NumPy`

🏆 2ND PLACE GLOBALLY • ASME IAM3D COMPETITION (40+ UNIVERSITIES)

Autonomous Payload Drone & Off-Terrain Rover

GitHub Code ↗

ASME UMass Amherst Mechatronics Team • Electrical & Prototyping Subteam Lead

Led electrical and prototyping subteam to engineer an articulated payload drone and off-terrain rover. Modeled custom CNC aluminum 6061 frame in SolidWorks, programmed ARM Cortex motor telemetry, and secured 2nd Place globally among 40+ international teams.

2nd Place	4.5 kg	50 Hz
Global ASME IAM3D Finish	Payload Capacity with FEA Tuning	Real-time Telemetry over SPI/I2C

`SolidWorks` `CNC Machining` `ARM Cortex` `Embedded C` `3D Printing` `I2C/SPI/UART` `Motor Drivers`

⚡ 100% DOE ELECTRICAL SAFETY COMPLIANCE

Offshore Wind Turbine 24V Power & Telemetry System

GitHub Code ↗

U.S. Department of Energy Collegiate Wind Competition • Electrical Subteam Lead (6 Engineers)

Led a 6-engineer subteam in modeling, simulating, and validating a 24V synchronous DC-DC power regulation circuit and sensor telemetry platform for dynamic wind-load conditions, achieving 100% safety and specification compliance.

24V ±0.5%	100%	6 Engineers
Voltage Regulation Stability	DOE Benchmark Compliance	Led Schematic & Assembly

`LTSpice` `Power Electronics` `24V DC-DC` `PCB Simulation` `Embedded C` `Telemetry`

Multi-Tenant Cloud Infrastructure & Terraform Automation

GitHub Code ↗

UMass Amherst IT • Cloud Infrastructure Engineering Intern

Built modular Terraform blueprints and automated CI/CD pipelines on AWS, accelerating developer onboarding and cloud environment provisioning by 40% with zero configuration drift across 6 engineering teams.

`AWS (EC2, S3, Lambda, IAM)` `Terraform` `Docker` `Python` `Bash` `CI/CD`

Embedded IoT Learning Platform

GitHub ↗

Stem Studio • Systems Lead

Created hands-on embedded firmware lab curriculum (ESP32, ARM Cortex, I2C/SPI) for 150+ students, raising project pass rates by 35%.

`Embedded C` `ARM` `ESP32` `IoT`

Malware Remediation & Hardening

GitHub ↗

Production Incident Response

Neutralized obfuscated PHP cloaking malware, restored WSOD failures, and implemented fail2ban and WAF security locks.

`Linux Hardening` `PHP` `Nginx` `Bash`